Lecture 27

Sequencing: Domains of Application

Bioinformatics Analysis Domains

A recap from Chapter 1. Bioinformatics projects can be divided into the following categories:

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1. Re-sequencing
2. Assembly
3. Classification
4. Quantification

And various combinations of the above.

Refer to the book

In this course we go into more detail on

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1. Re-Sequencing
4. Quantification

In this lecture however we will provide you with a high level overview of other topics.

Refer to the **Biostar Handbook** for code examples.

1. Re-sequencing recap

Re-sequencing a known genome

We have a known genone: **reference** genome. Obtain sample from a "variant" genome.

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1. Compare against reference using **alignments**.

2. From **alignments** produce **variants**.

3. **Variants** have **effects** on genomic products.

4. Tabulate the variants based on their effects on **phenotypes**

2. Assembly

Genome Assembly

Genome is unknown or only remotely similar to another known genome.

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1. Assemble reads into longer sequences called **contigs**.

2. **Scaffolding**. Orient **contigs** relative to one another.

3. **Gap closing**. Attempt to fill in the **gaps** between the oriented scaffolds. (Done experimentally + computationally: trial-end-error).

Assembly: Contig Building

Requirement: genome assembly requires sufficient coverage so that the **ends of reads** overlap.

```
---- ----              -----  ----- -----
  -----                  ------- -----
```

From this overlap the **assembler** produces **consensus** sequences.

```
---------                  (from the left side)
------------------         (from the right side)
```

Each single **contig** is a superposition of many reads.

Assembly: Contif Scaffolding

If we had a related sequence we could align

```
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         ---------          ----------
```

From that we would could orient the contigs:

```
         ---------NN 13 bases NN----------
```

or even fill in all bases

```
         -------------------------
```

Assembly: Gap Closing

Once **contigs** are oriented into **scaffolds**

```
---------NNNNNNN-----------
```

We need a way to bridge over the `N`s and establish the sequence identiy. Needs a new experiment.

```
---------NNNNNNN---------

    ---     ----
      --- --- -----
        --- -----
```

Does not need high throughput sequencing!

Assembly challenges

Genome assembly is the least robust and most **neglected** field of bioinformatics.

1. Algorithms are extremely sensitive to parameter settings

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2. Most assemblers feel like **black boxes**. We don't understand the decisions that the software makes.

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3. Most assemblers make "questionable" decisions all the time. Lots of head-scratching moments.

Assembly as a field is in a underdeveloped state because most life scientists do not understand its value.

Assembly is an attempt to produce the "real" sequence.

Not a variant relative to something else.

That's why it is so hard to get it right.

A Personal Opinion

In my opinion: Genome Assembly is the **most important** field of bioinformatics.

We should stop treating biological data in terms of **reference** and **variant**. After all everything is a **variant** of some kind.

It is ineffective and counterproductive to designate one of the **variants** as the **reference**. Then discuss all other variants relative to this **reference** variant. Yet that is exactly what we currently do.

We need a conceptual change in the field.

3. Classification

Classification: What's in my pot?

Typically refers to establishing the composition of a diverse population based on DNA fragments obtained from that population.

Instead of one kind of DNA you could have hundreds, thousands, tens of thousands or more. You have to deal with all kinds of complications caused by:

1. Different abundances of each species.
2. Different genome length of each species.
3. Regions of high similarity between species.

How many species per sample?

How diverse is a sample?

How many different species would you expect from the "most" diverse DNA extract from one location: soil, gut, lake bottom etc

Typical species richness

1. Gut: **300-1000** species (**99%** of the bacteria come from about **30** or **40** species)
2. Soil: **10,000** species per sample. Only about **500** of which would be "culturable" in a laboratory.

Is there a catch?

It is biology my friends.

There is always a catch 😉

Different species for everyone

Gut: **300-1000** species (**99%** of the bacteria come from about **30** or **40** species)

But these are not the same species for everyone.

The most abundant bacteria from one gut may not even be present at all in another gut.

It is not about the species. It is about the function that a bacterial provides. Different species may provide similar functions.

How many total bacterial species?

Not quite clear.

Further complicated by a lack of definition of what **species** should mean when it comes to bacteria.

Classical (Linnean) taxonomy (that defines the word species) was meant to desribe higher order organisms that evolve top-down (parent --> child).

It does not work well for bacteria that can evolve laterally (it can incorporate DNA from another unrelated bacteria).

Classification: Using marker genes

You can simplify some complexities by using a "marker" gene for the species of interest.

Isolate and sequence only the marker gene.

Since the marker gene is usually short enough and specific enough it simplifies the problem to only:

1. Different abundances of each species.
2. **SOLVED**: Different genome length of each species.
3. **SOLVED**: Regions of high similarity between species.

Classification: 16S ribosomal RNA

Has a slow rates of evolution - plus the gene is so specialized and important that every bacterial species has it and could not replace it.

Entire subfield of bioinformatics: 16S Classification.

Classfiers are fast and efficent. Do not require substantial computational power.

Classifiers produce a file with read counts per taxonomical level.

Requires a statistical methods to detect which counts stayed the same.

Classification: 16S results

Classification is perfomed relative to a taxonomy.

taxid	name	rank	Sample1	Sample2	Sample
0	Root	rootrank	7280	3590	8891
1	Bacteria	domain	7256	3572	8850
841	Proteobacteria	phylum	2674	3055	933
1501	Gammaproteobacteria	class	2653	3039	933
1616	Enterobacteriales	order	2589	3030	933

Reports will include the classification at each taxonomy level.

Some sequences may only be classified at a higher level: We get to know the phyla but not the order.

Whole genome classification

Marker genomes don't tell us about function. Yet that may be the most important characteristic.

So how do we characterize the entire genomes of an entire population.

We have to deal with:

1. Different abundances of each species.
2. Different genome length of each species.
3. Regions of high similarity between species.

Understand the process

It is important that you understand the sequencing concepts.

1. DNA is extracted from each species
2. DNA is fragmented

Longer bacteria produce more fragments.

Abundant bacteria produce more fragments.

The sequencer measures some fragments, but not all.

The rules of classification

You never know the **absolute** number of species - only the **relative** abundances.

Think of it this way.

Imagine that you are counting animals in a forest. But you can only count up to 10 and whatever animal you run into you have to count it.

Forest 1: 5 bears, 5 squirrels
Forest 2: 2 bears, 4 rabbits, 2 bobcats, 2 squirrels

Does **Forest 1** have more bears than **Forest 2**?

Interpreting classification: relative abundances

Forest 1: 5 bears, 5 squirrels
Forest 2: 2 bears, 4 rabbits, 2 bobcats, 2 squirrels

Does **Forest 1** (5 bears) have more bears than **Forest 2** (2 bears)?

We don't know. Maybe we had to scour the entire **Forest 1** to find 5 bears, whereas we found 2 bears after spending a minute of **Forest 2**.

What we do know is that ratio of bears to squirrels is the same in both! In addition we have other species in **Forest 2** with twice as many rabbits as bears.

Metagenome assembly

Ever regular genome assembly has many challenges.

Producing a single well assembled genome is a major undertaking.

How well do you think it will work when we try assemble thousands of possibly similar genomes. There you have your answer.

If bioinformatics is a **toddler** in a sandbox, then metagenome assembly is like a **six week old baby**. Makes a lot of fuss and is tedious and tiring to deal with.

4. Quantification: See next lecture.